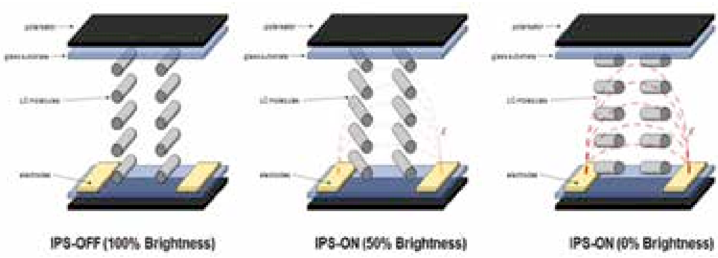
There is one more drawback of TN panels, i.e. they are only capable of representing 75b per cent of the colour gamut out of the 24-bit available from the graphics card which translates to roughly 6-bits for the each colour of RGB. In the meanwhile IPS displays can reproduce images with 8-bit colour depth per sub-pixel. This means images viewed on an IPS panel will seem more like real world colours.

The working of IPS panel

Before delving into the specifics of how an IPS display works, it's essential to know the working mechanism of a TFT LCD which is almost similar for all monitors. Refer previous chapters to know more.

The panel design of TN and IPS panels differ greatly as both the electrodes are present on the same wafer in IPS panels whereas in TN panels it's present in two different layers. This makes the response time greater than TN panels. In general, response time is the time taken for a pixel to turn from one shade of grey to another when a voltage is applied to the said pixel. There is no industry standard so grey to grey values are measured differently by different manufacturers. So in IPS panels, this arrangement of electrodes increases the response time of the panel but ensures better colour reproduction.



Brightness control in IPS panels

In TN panels, the pixels or liquid crystals (LCs) are also anchored perpendicular to the glass substrate, while in the case of IPS panels, the LCs are anchored parallel to the substrate. So this horizontal arrangement led to the use of a pair of transistors to maintain the parallel positions of the LCs in the displays when in the 'on' state. This same arrangement helps in changing the brightness of the panel by adjusting the angle of LCs to 0, 45 or 89 degrees which in turn ensures better contrast ratios.